Process Note: Retrieval of Crop Monitoring Indices Using Copernicus Data Space Browser

Objective:

To extract standardized vegetation, moisture, thermal and structural indicators for selected farm polygons (KMLs) using European Space Agency satellite datasets available through the Copernicus Data Space platform.

Platform & Data Source:

1. Platform: Copernicus Data Space Browser
2. Primary Satellites Used: Sentinel-2 (Optical) – Vegetation & Moisture Indices, Sentinel-3 – Land Surface Temperature (LST), Sentinel-1 (Radar) – Structural Biomass (SAR)

Retrieval Mechanism:

1. Sentinel-2 L2A

a. Dataset Selection: Sentinel-2, Level-2A (Surface Reflectance), Cloud filter ≤ 20%

b. Time Window: Select image corresponding to defined crop stage (e.g., 30 DAS, 60 DAS)

c. Indices having direct retrieval option:

1. NDVI - Normalized Difference Vegetation Index
2. SAVI – Soil Adjusted Vegetation Index
3. NDMI - Normalized Difference Moisture Index

Steps for downloading the mean value for these indices:

- Select index layer under Agriculture/Vegetation theme.
- Open “Statistical Information”.
- Record mean value for selected farm polygon.
- Export or manually tabulate values.

d. Indices that need to be derived with manual operation:

4. MSAVI – Modified Soil Adjusted Vegetation Index
5. NDRE - Normalized Difference Red Edge
6. GCI - Green Chlorophyll Index
7. PSRI - Plant Senescence Reflectance Index
8. LAI – Leaf Area Index

Steps for calculating the mean value for these derived indices:

- Open “Spectral Explorer – Values”.
- Retrieve mean reflectance for required bands:

- B03 (Green)
- B04 (Red)
- B05 (Red Edge)
- B08 (NIR)
- B11 (SWIR)
- B02 (Blue Band)

- Retrieve Apply standard mathematical formula in spreadsheet:
  For example, NDRE = (NIR − RedEdge) / (NIR + RedEdge)
  GCI = (NIR / Green) – 1
  PSRI = (Red − Green) / NIR
  MSAVI = $(2NIR + 1 - \sqrt{((2NIR + 1)^2 - 8(NIR - Red))}) / 2$
  LAI (Leaf Area Index) = 3.618 × EVI - 0.118
  Where EVI (Enhanced Vegetation Index) is:
  EVI = 2.5 × (NIR - Red) / (NIR + 6×Red - 7.5×Blue + 1)
  [Note: This index is calculated using custom EvalScript on the Copernicus browser. This uses the EVI-based empirical relationship.This formula uses: **B08** (NIR - Near Infrared band): Highly reflected by healthy vegetation, **B04** (Red band): Absorbed by chlorophyll in vegetation, **B02** (Blue band): Helps reduce atmospheric effects]

2. Sentinel-3 SLSTR L1B

a. Dataset Selection: Sentinel-3 SLSTR L1B , Cloud filter: ≤ 20%

b. Time Window: Select image corresponding to defined crop stage

c. Indices that need to be derived with manual operation:

1. LST (Land Surface Temperature) – This index is calculated using custom script on the Copernicus browser that runs Sobrino et al. (2004) Split Window Algorithm, which relies on two thermal infrared bands from SLSTR:

Band	**Wavelength**	**Measurement**
S8	10.85 µm	Brightness Temperature (K)
S9	12.0 µm	Brightness Temperature (K)

$$LST(K) = T8 + 1.06 \times (T8-T9) + 0.46 \times (T8-T9)^2 + 54.3 \times (1 - \varepsilon)$$

$$LST(°C) = LST(K) - 273.15$$

Where:

- **T8** = Brightness Temperature at S8 (Kelvin)

- **T9** = Brightness Temperature at S9 (Kelvin)
- **ε** = Land surface emissivity (~0.97 for general land)
- The term 54.3 × (1 − ε) corrects for emissivity deviation from a blackbody

Note: Custom EvalScript for Split window algorithm. This needs to be run in the Copernicus Browser:

```javascript
javascript
//VERSION=3
//LST (Land Surface Temperature) - Split Window Algorithm
//Dataset: Sentinel-3 SLSTR L1B
//Bands used: S8 (10.85 um) and S9 (12.0 um) - Brightness Temperature in Kelvin
//Algorithm: Sobrino et al. (2004) Split Window
//Color scale: Blue=0°C, Green=20°C, Yellow=35°C, Red=50°C+

function setup() {
 return {
  input: ["S8", "S9", "dataMask"],
  output: { bands: 4 }
 };
}

function evaluatePixel(sample) {
 var T8 = sample.S8;   // Brightness Temperature at S8 (10.85 μm) in Kelvin
 var T9 = sample.S9;   // Brightness Temperature at S9 (12.0 μm) in Kelvin

 var epsilon = 0.97;   // Fixed land emissivity
 var dT = T8 - T9;     // Split window difference

 var LST_K = T8 + 1.06 * dT + 0.46 * dT * dT + 54.3 * (1 - epsilon);
 var LST_C = LST_K - 273.15;  // Convert to Celsius

 // Color ramp: 0°C=blue, 20°C=green, 35°C=yellow, 50°C+=red
 var t = Math.max(0, Math.min(1, (LST_C - 0) / 50.0));
 var r, g, b;
 if (t < 0.25)      { r = 0; g = t * 4; b = 1; }
 else if (t < 0.5)  { r = 0; g = 1; b = 1 - (t - 0.25) * 4; }
 else if (t < 0.75) { r = (t - 0.5) * 4; g = 1; b = 0; }
 else               { r = 1; g = 1 - (t - 0.75) * 4; b = 0; }

 return [r, g, b, sample.dataMask];
}
```

As a result, The LST map will now be live on the browser:

- **Urban area** shows **yellow-orange-red** = high LST (35–50°C), typical urban heat island

- **Water bodies** (reservoirs) show **blue** = low LST (~15–20°C)
- **Vegetation/forests** show **green** = moderate LST (~20–30°C)
- **Bare soil/open land** shows **orange-red** = high LST (40°C+)